

CredRisk.AI

A Loan Intake Portal and underwriter workbench for the full SME to analyst to SME credit journey.

SME-owned intake

Random Forest scoring

AI output gate

Published SME outcome

SME

Detailed application, five-year plan, and evidence upload start on the applicant side.

RF

One governed Random Forest baseline produces the 0-1 application risk score.

A-F

Analysts publish a reviewed rating and applicant-safe report back to the SME.

SME underwriting breaks when intake, evidence, scoring, and publication sit in separate tools.

The bank needs speed, but cannot lose traceability.

Applicant inputs, uploaded evidence, fraud signals, AI output, model grades, and human decisions must remain connected without letting automation make the final call.

Incomplete intake

SMEs need one place to enter company context, financials, loan purpose, and the five-year plan.

Evidence friction

Missing, ambiguous, risky, clean, and fraudulent document cases should be visible before review.

Governance risk

Model score, AI advice, analyst rating, and SME-facing publication need separate controls.

CredRisk.AI turns the loan file into a controlled end-to-end workflow.

1

Intake

The SME logs in with DemoUser, uses SME SSO, fills the detailed Loan Intake Portal, and submits evidence.

2

Score

The lender opens a frozen snapshot and applies the single Random Forest risk model.

3

Review

Case Review is gated behind AI Output, so the analyst sees the generated package first.

4

Publish

The SME receives only the lender-approved rating, message, and applicant-safe report.

CredRisk.AI gives lenders faster decisions without hiding who decided what.

Better intake

SMEs can load sample cases or start blank, with help pop-ups explaining unfamiliar fields.

Evidence trust

Sample document packs cover clean, neutral, risky, fraudulent, and ambiguous scenarios.

Explainable score

Risk drivers, ratios, and interpretation columns translate model signals into review language.

Human accountability

Ms. Cooper records the analyst action and rating separately from the Random Forest grade.

AI where useful

The AI package creates a private lender report and applicant-safe SME draft before review.

Clean operations

Submitted SME cases stay in Home and Personal Workspace; synthetic queue work stays in Operations Desk.

The MVP maps the business plan into working software, not a mockup.

1

Role-aware Streamlit UI

SME Loan Intake Portal, lender Home, Personal Workspace, Operations Desk, dashboards, settings, tutorials, and support.

2

Data and evidence pipeline

Synthetic SME portfolio, realistic sample companies, local document vault, hashes, and generated sample evidence.

3

Random Forest risk engine

Scikit-learn preprocessing, derived credit and fraud signals, 0-1 risk score, A-F grading, and model metrics.

4

AI report layer

Deterministic fallback plus OpenAI or local LLM path for internal and applicant-safe report generation.

5

Runtime state and audit trail

Submitted snapshots, reviews, AI outputs, validation results, publications, and demo persistence stay traceable.

The live demo follows the pitch script: SME to analyst to SME.

1

SME login

Choose SME company, use DemoUser and SME SSO, then open the Loan Intake Portal.

2

Loan Intake Portal

Load a sample evidence case or blank intake, complete the detailed form, attach documents, and submit.

3

Lender handoff

Log in with YourBank SSO; Home shows Suggested Actions and the SME Portal Intake snapshot.

4

Personal Workspace

Open the locked intake, inspect evidence, view the Random Forest score, and run Risk Analysis.

5

AI Output and Case Review

Generate the internal and SME reports, then record the analyst decision and publish the approved copy.

6

SME outcome

Return as the SME to see the published rating, lender message, and downloadable evaluation report.

The MVP is focused; production needs integrations, controls, and monitoring.

Scale prerequisites

PSD2, accounting, registry/KYB, document ingestion, role-based access, secure secrets, deployment controls, and model monitoring.

PSD2

Accounting APIs

KYB registry

Document review

Managed risks

False positives, fraud misses, AI hallucination, data leakage, model drift, and over-reliance on automation are handled through review gates and auditability.

Human final decision

AI review gate

Validation metrics

Audit trail

The repository is built for the FinTech assignment deliverables.

Working MVP

The app demonstrates the full value loop: intake, evidence, scoring, AI package, review, publication, and monitoring.

Clean repository

README, demo script, requirements, Windows launcher, commented Python files, and research notes support handoff.

Agent orchestration

.codex/CODEX.md documents page structure, workflows, acceptance checks, and agent guidance for repo-aware changes.

CredRisk.AI

Loan Intake Portal
Underwriter Workbench

NOW THE DEMO

CredRisk.AI turns SME lending from a file chase into an explainable decision workflow.

The live walkthrough shows Loan Intake Portal, frozen lender review, Random Forest scoring, AI Output, human publication, and the SME-facing result.